



Improving Definition Modeling via Harmonizing Semantic Experts

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Motivation: The Challenges of Definition Modeling

Definition Modeling (DM) faces inherent bottlenecks in current LLMs:

- ▶ **Polysemanticity:** Difficulty capturing rare word senses and subtle distinctions.
- ▶ **Superposition:** Dense architectures force highly polysemantic representations to share neurons, causing interference.
- ▶ **Specificity:** LLMs' generations are under-generic or overly specific, even with In-context Learning.



Figure 1: Examples of context-dependent definitions (e.g., Space Needle, Julie Delpy, Stratosphere, and Genderqueer).

Methodology: The Specialize-then-Synthesize Framework

LM-LEXICON introduces an effective hybrid training approach:

1. **Data Partitioning:** Clustering data into semantic shards using embeddings.
2. **Expert Training:** Asynchronous training of small LMs as domain experts.
3. **Model Merging:** Composing expert weights into a unified MoE.
4. **Semantic Routing:** Proposing a semantic-aware domain-level routing mechanism.

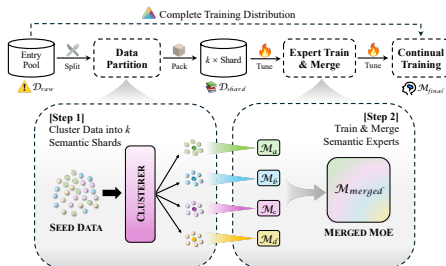


Figure 2: The Specialize-then-Synthesize pipeline of LM-LEXICON.

Technical Details: Model Merging & Semantic Routing

In  LM-LEXICON, we merge feed-forward networks (FFNs) as standard mixture-of-expert modules, while applying algebraic averaging to other parameters:

$$\text{FFN}_{\text{MoE}}^{\ell}(x) = \sum_{i=1}^N \mathcal{G}(x) \cdot \text{FFN}_i^{\ell}(x)$$

Unlike token-level routing, we use **domain-level sequence routing**:

- ▶ Input representations are routed to the nearest cluster centroid via cosine similarity.
- ▶ **Result:** Higher efficacy (+1%) than trivial token-level routing for DM tasks.

Datasets & Statistics

Benchmark Datasets:

- ▶ **WordNet:** Online lexical database with terms & definitions.
- ▶ **Oxford:** Built from Oxford Dictionary entries.
- ▶ **Wikipedia:** Tests phrase descriptions beyond single words.
- ▶ **Urban:** Internet slang and urban vocabulary.
- ▶ **3D-EX:** Largest English DM dataset, aggregates all above resources.

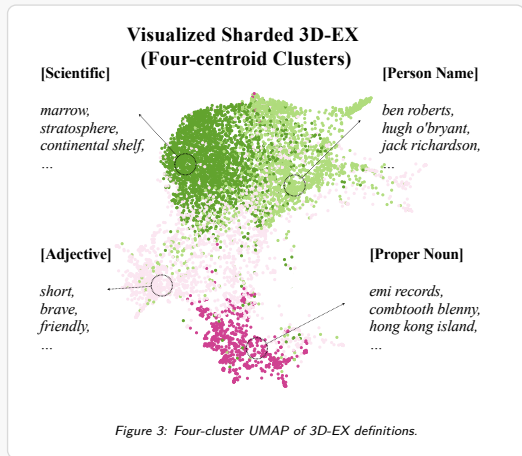
	WordNet	Oxford	Wikipedia	Urban	3D-EX
genre	formal	formal	web	idiom	misc.
domain	synset	lexicon	encyclopedia	slang	multi
publish year	2017	2018	2018	2017	2023
$\# S_{train}^t$	13,883	97,855	887,455	411,384	1,309,312
$\# S_{valid}^t$	1,752	12,232	44,003	57,883	513,789
$\# S_{test}^t$	1,775	12,232	57,232	36,450	450,078
# glo. per term	1.75 ± 1.19	2.99 ± 4.41	5.86 ± 78.25	2.11 ± 2.92	6.00 ± 53.78
# tok. per term	1.00 ± 0.00	1.00 ± 0.00	1.85 ± 0.93	1.44 ± 0.72	1.45 ± 0.78
# tok. per ctx.	5.79 ± 3.44	19.02 ± 9.18	19.68 ± 6.31	11.36 ± 6.02	18.82 ± 9.99
# tok. per glo.	6.64 ± 3.78	11.41 ± 7.13	5.97 ± 4.51	11.02 ± 6.86	8.97 ± 6.76
% overlap rate	0.00 / 0.00	80.72 / 0.09	0.00 / 0.00	20.62 / 20.56	0.00 / 0.00

Table 1: Dataset statistics and lexical overlap.

Key Statistics:

- ▶ Mean/std of per-term, per-context, per-gloss metrics reported.
- ▶ **Lexical overlap:** $|S_{train}^t \cap S_{test}^t| / |S_{test}^t|$
- ▶ Clustering performed on 3D-EX for semantic expert finetuning.

Semantic Clustering & Experts Specialization



The training corpora (e.g., 3D-EX) is clustered into 4 distinct semantic domains:

- ▶ **Scientific:** Technical definitions (e.g., *Stratosphere*).
- ▶ **Adjective:** Descriptive terms (e.g., *Brave*).
- ▶ **Person Name:** Biographical data (e.g., Ben Roberts).
- ▶ **Proper Noun:** Entities and locations.

Impact: Enables fine-grained experts specialization (+10% quality improvement).

Main Results: Comparison with Prior SoTA

Model	WordNet	Oxford	Wiki	Urban	3D-EX	Avg.
Rerank-T5	30.9	28.0	57.2	18.2	38.5	34.6
GPT-4-Turbo	29.7	34.3	40.6	26.5	37.6	33.7
Claude-3-Opus	21.7	25.9	15.8	7.1	24.6	19.0
LM-Lexicon (4×8B)	40.6	33.7	57.6	33.8	46.3	42.3

Table 2: Performance comparison (ROUGE-L).  LM-Lexicon achieves ~7–10% improvement.

- ▶ **Outperformance:** Beats supervised baselines and frontier LLMs.
- ▶ **Robustness on Rare Words:** Exceptional on Urban Dictionary (slang/idioms).

Human Evaluation & Qualitative Analyses

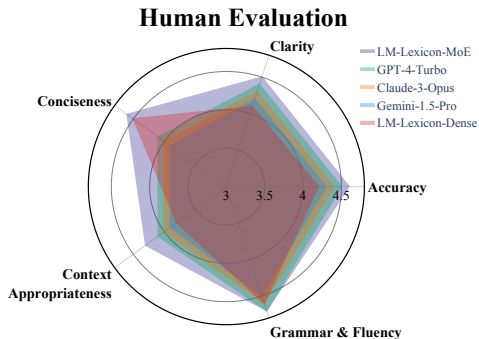


Figure 4: Human evaluation across 5 criteria.

Evaluated by linguistics experts on 5 criteria:

- ▶ **Accuracy:** Superior at capturing the core meaning.
- ▶ **Clarity:** Outperforms GPT-4/Claude-3, which are often overly verbose.
- ▶ **Context Appropriateness:** Better adherence to pragmatic constraints.

WIC: "I'll be a gone coon when the battle starts"

Reference: "an eccentric or undignified rustic"

GPT-4: [Hallucinates about raccoons or slurs]

Ours: "A person who is deemed to be despicable".

Ablation Studies

Model	BLEU	ROUGE	p-value
 LM-LEXICON	45.69 $\pm$ 0.3	46.07 $\pm$ 0.1	—
+ w/ no split	35.13 $\pm$ 0.2	43.46 $\pm$ 0.3	2.9e ⁻⁵
+ w/ random split	36.24 $\pm$ 1.4	43.58 $\pm$ 0.8	1.6e ⁻⁵
+ w/ lexical split	38.13 $\pm$ 0.5	44.12 $\pm$ 0.6	1.3e ⁻⁴

Table 3: Data partition method comparison.

Data Partition Methods:

- ▶ **Semantic clustering:** Outperforms lexical (TF-IDF) by +7% BLEU, +1% ROUGE.
- ▶ Semantic-targeted clusters capture more precise senses.
- ▶ Enables robust experts across domains.

Model	BLEU	ROUGE	p-value
 LM-LEXICON	45.69 $\pm$ 0.3	46.07 $\pm$ 0.1	—
+ w/ top-1 token-level	43.12 $\pm$ 0.4	43.79 $\pm$ 0.5	1.9e ⁻³
+ w/ top-2 token-level	45.38 $\pm$ 0.2	45.21 $\pm$ 0.1	8.6e ⁻¹
+ w/ sequence-level	44.47 $\pm$ 0.2	44.82 $\pm$ 0.3	2.7e ⁻³

Table 4: Routing policy comparison.

Routing Policies:

- ▶ **Domain-level routing:** Most effective approach.
- ▶ Surpasses popular top-2 token-level routing.
- ▶ Semantic routing via domain clusters benefits semantic-intensive tasks.

Scalability of Semantic Experts

Scaling Semantic Experts

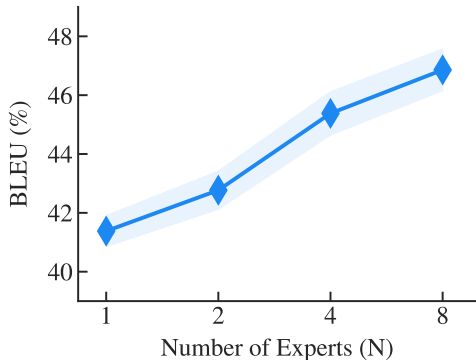


Figure 5: Scaling test performance on 3D-EX, with varying number of experts.

Investigated impact of number of semantic experts ($N = 1, 2, 4, 8$):

- ▶ $N = 1$: Collapses to dense model (baseline).
- ▶ $N > 1$: Expands to sparse MoE architecture.
- ▶ **Scalability:** Performance consistently increases with more experts.

Bleu Performance:

$N = 1$: 41.38%; $N = 8$: 46.86%

Trend suggests potential for even finer-grained semantic experts.

Impact of Test-time Scaling

Performance vs. Repeated Sampling Scale

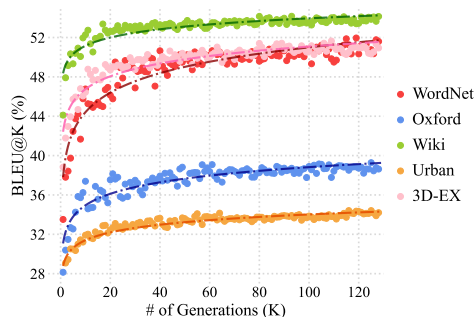


Figure 6: Performance with repeated sampling (Oracle verifier).

Two verification strategies tested:

- ▶ **Oracle:** Reference-based binary feedback.
- ▶ **BoN-ORM:** Outcome reward model trained for providing scalar feedback.

Key Findings:

- ▶ Oracle verifier: $\Delta_{BLEU} > 2\%$ for Dense model
- ▶ Optimal: 128 samples with Oracle
- ▶ Consistent gains across benchmarks with both verifiers

Conclusion

- ▶ **Effective Framework:** Specialize-then-Synthesize successfully harmonizes semantic heterogeneity, thereby enhancing the DM performance of  LM-Lexicon.
- ▶ **State-of-the-Art:** Significant gains (+10%) on definition modeling benchmarks.
- ▶ **Scalability:** Performance scales linearly with the number of semantic experts.

Thank You !



Project Page: <https://lm-lexicon.github.io>



Data and Models: <https://huggingface.co/LM-Lexicon>